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## **EX-703-GS**

### **B.E. VII Semester**

Examination, December 2020

### **Grading System (GS)**

### **Digital Signal Processing**

*Time : Three Hours*

*Maximum Marks : 70*

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. State and prove following properties of DFT.
  - i) Circular Frequency Shift
  - ii) Periodicity
2. Determine whether the following system are linear, time-invariant and causal:
  - i)  $y(n) = x(n) \cos x(n)$
  - ii)  $y(n) = x(-n+2)$
  - iii)  $y(n) = n x^2(n)$
3. Compute the response of the system:  
 $y(n)=0.7y(n-1)-0.12y(n-2)+x(n-1)+x(n-2)$   
to the input  $x(n) = n u(n)$ . Is the system stable.
4. a) Obtain the Fourier transform of:  
 $x(t) = 5[u(t+3) = u(t+2) - u(t-2) - u(t-3)]$   
b) Write the applications of FFT. State its computational requirements.

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5. Obtain the cascade form realizations of FIR systems

$$H(z) = 1 + 5/2 Z^{-1} + 2 Z^{-2} + 2 Z^{-3}$$

6. a) Convert the analog filter into a digital filter whose system

function is  $H(s) = \frac{s + 0.2}{(s + 0.2)^2 + 9}$ . Use impulse invariant technique. Assume  $T = 1s$ .

- b) Discuss the realization of IIR filters in canonic and parallel forms.

7. Explain the following window techniques for designing FIR filters.

- a) Kaiser window
- b) Hamming window

8. Write short notes on (any two):

- i) One sided Z-transform
- ii) FIR Digital Filters Characteristics
- iii) Bilinear Transformation Method

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